

MILITARY APPLICATIONS OF UAVS



The DRDO is actively developing a number of new drones and associated payloads, to enhance the defensive capabilities of the armed forces

Aditya Madanapalle | aditya@digit.in

Drones have a number of uses for the military, and DRDO is exploring all of them, with a focus on increasing self reliance.

This means that the systems and subsystems are developed and manufactured in India itself, to whatever extent is possible. The drones are for surveillance, monitoring coastlines, anti-piracy measures, supporting the goals of the infantry, and as target practice for other weapons systems. One of the most cutting edge applications combines drones and virtual reality to provide training for situations that would be impossible to recreate in real life, without exposing the personnel to significant risks. This training regime is enabled by a computer game developer called Threye.

LIVE, VIRTUAL, CONSTRUCTIVE SIMULATION

This operation was conducted to test out the potential and capabilities of an Live, Virtual and Constructive (LVC) simulation for defence purposes. There are many situations that are just too expensive or hazardous to simulate – such as nuclear, biological or

chemical warfare. At times, a unit may require first hand tactical knowledge of an area, before getting deployed on the ground. The LVC simulation comes into play here.

The idea is simple. A drone is dispatched to survey a particular area. It flies around, and captures the area with stereoscopic images. A depth map is obtained from this drone, which is then populated with digital assets to represent the actual real world analogues. These can include vegetation,

buildings, boards, vehicles and so on. The personnel get trained in a virtual environment, wearing VR headsets. The fidelity is on a meter scale, that is, what the personnel can see from their eyes is pretty close to what they would actually see on the field. This fidelity is about 30 percent better than what is offered by conventional services, such as Google Maps. Threye demonstrated such a simulation using off the shelf parts, a DJI quadcopter, a GoPro camera, and the Unity game engine. The



A high resolution virtual environment using off the shelf quadcopters and cameras